

# ARM Cortex-M Debug & Fault System - Quick Cheat Sheet

| Register | Address    | Purpose                                  | Key Bits / Meaning                                     |
|----------|------------|------------------------------------------|--------------------------------------------------------|
| CFSR     | 0xE000ED28 | MemManage + BusFault + UsageFault status | [31:16] UFSR, [15:8] BFSR, [7:0] MMFSR                 |
| HFSR     | 0xE000ED2C | HardFault status                         | FORCED(30), VECTTBL(1)                                 |
| DFSR     | 0xE000ED30 | Debug fault status                       | EXTERNAL(4), VCATCH(3), DWTTRAP(2), BKPT(1), HALTED(0) |
| MMFAR    | 0xE000ED34 | MemManage fault address                  | MMARVALID (CFSR[7])                                    |
| BFAR     | 0xE000ED38 | BusFault fault address                   | BFARVALID (CFSR[15])                                   |
| AFSR     | 0xE000ED3C | Auxiliary fault info                     | Implementation defined                                 |

| Fault Type | Bit | Flag        | Meaning                      |
|------------|-----|-------------|------------------------------|
| MemManage  | 0   | IACCVIOL    | Instruction access violation |
|            | 1   | DACCVIOL    | Data access violation        |
|            | 3   | MUNSTKERR   | Error during unstack         |
|            | 4   | MSTKERR     | Error during stack           |
|            | 5   | MLSPERR     | Lazy FP error                |
|            | 7   | MMARVALID   | MMFAR address valid          |
| BusFault   | 8   | IBUSERR     | Instruction bus error        |
|            | 9   | PRECISERR   | Precise data bus error       |
|            | 10  | IMPRECISERR | Imprecise data bus error     |
|            | 11  | UNSTKERR    | Bus error during unstack     |
|            | 12  | STKERR      | Bus error during stack       |
|            | 15  | BFARVALID   | BFAR address valid           |
| UsageFault | 16  | UNDEFINSTR  | Undefined instruction        |
|            | 17  | INVSTATE    | Invalid state                |
|            | 18  | INVPC       | Invalid PC load              |
|            | 19  | NOCP        | No coprocessor               |
|            | 24  | UNALIGNED   | Unaligned access             |
|            | 25  | DIVBYZERO   | Divide by zero               |

| Register | Purpose                         | Key Bits                                    |
|----------|---------------------------------|---------------------------------------------|
| DHCSR    | Debug halting control/status    | C_DEBUGEN(0), C_HALT(1), S_HALT(16)         |
| DEMCR    | Debug exception/monitor control | TRCENA(24), VC_HARDERR(16), VC_CORERESET(0) |
| AIRCR    | Reset & priority grouping       | VECTKEY(31:16), SYSRESETREQ(2)              |